Next-Generation Power & Cooling Solutions

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Navigating Industry Transformation

End of 2021: Challenges at the Start

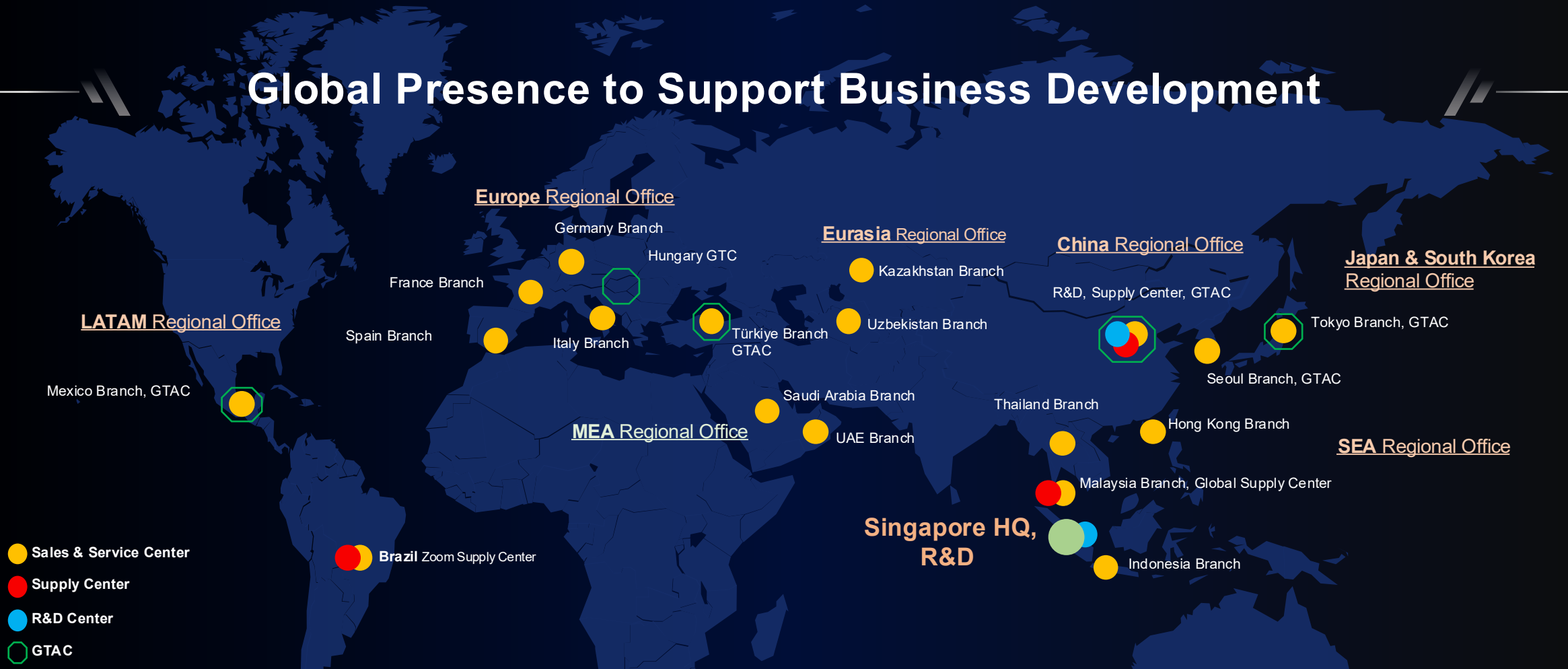
1. **Single product focus**: failing to seize major market opportunities, such as the 4 billion+ market for Internet servers, 3 billion+ for GPU servers (USD)
2. **Lack of industry/customer case studies**: no established use cases or proven success in the industry
3. **Insufficient qualifications**: missing foundational certifications and credentials

End of 2025

1. **Computing Infrastructure and Services**
   - **Global: 60%+ CAGR from 23-25; Ranked 6th overall** (IDC Report 2025)
   - **China**: **Ranked 2nd overall**, with **first-place positions** in the AI solution, liquid-cooled data centers and AI Appliance(IDC Report 2025)
2. **Two New Businesses Launched a Year Early**
   - **Sapient Enterprises Ecosystem:** secured a milestone project – digital transformation of core operations for large enterprises.
   - **Smart Energy Solution:** more than one million contracts have been bid



Global Presence to Support Business Development



Global Reach

- 7 regional offices with local branches to provide localized service and support worldwide.
- 6 Global Technical Assistance Centers (GTACs) serving 10,000+ customers

Innovation Hubs

- Worldwide research & development centers are strategically located in key technology hubs

Supply Chain

- Wholly owned Global Supply Center in Malaysia
- A joint venture facility in Brazil

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AI Defines the Next Era, Fast Movers Lead

75%–100%

Google: 25% to 75% in 18 months

Anthropic: 100% AI-generated code

5 year -> Months

Drug discovery: Hit-to-Lead

AlphaFold 3 + Isomorphic
Lilly/Novartis already signed

1 wk -> 1 hr

30%+ of traditional subtitling firms shut down

AI recognition, translation, and alignment
Accuracy: 97%

3x Productivity

Foxconn "Dark factory"

AI-powered workflow, 3x productivity, near-zero human supervision

$250K

NVIDIA Token/Engineer/year

GTC 2026 · Jensen Huang: A $500k engineers should use $250k in tokens.

20 min → 3–5 sec

Accuracy: 98.2%

Tuberculoisis AI screening

2025 Tongji Hospital + Xi'an Northern Hospital

108-Language Real-Time Translation

MIIS: admissions suspended

Google translation ≈ native speaker

……

Reconstruction 01

Product & R&D

AI coding: 30%+ code generation
AI for Science
Drug discovery: 5-year → 18 months
New materials/simulation design

Reconstruction 02

Creation & Collaboration

Marketing · PPT
Strategy analysis · reports
Days → hours
AI drafting + manual finalization

Reconstruction 03

Operations & Decision-making

+ Token economy
Enterprise software revalued
License → Token
Job reshaping

Entering 04

Physical world

L4 autonomous driving
Humanoid robot
Surgical robots + physical AI

Source: KPMG Global AI Pulse, Q1 2026 (April 2026, 2,110 senior leaders)

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WATT→FLOPS→TOKENS→AGENTS→VALUES

Tokens: The Cost Center

Token as the basic unit of AI inference and decision cost
Computing resource consumption measured by —
Token Per Watt (TPW)



Agents: The Profit Center

Agent as the entity that delivers and earns
Computing value measured by —
Daily Active Agent (DAA) and Tasks Per Day (TPD)

Old Measures Cannot Gauge the New Value

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WATT→FLOPS: Three Revolutions, All At Once

01 Cooling Revolution

Extreme heat
GPU power → 3,000 W+

Liquid cooling + Diamond copper

Standardized liquid cooling	CO/HBM4/PCIe 6.0 drive broader adoption
45°C warm-water cooling	Free cooling and a PUE of 1.05
Diamond plate	Thermal conductivity 1,500+W/m·K
Waste heat recovery	60°C return water for heating and industrial preheating

02 Power Revolution

Energy surges + Supply bottleneck
300 kW per rack · IR drop: 15–20%

HVDC + VPD

800 V HVDC power shelf	300 kW per rack, fewer conversions
VPD	90% shorter power path from VRMs beneath the chip
Rack-level BBU	Controlling di/dt for 3,000 W+ chip
SiC / GaN / Ga_2O_3	Wide-bandgap semiconductors reaching 98%+ efficiency

03 Interconnect Revolution

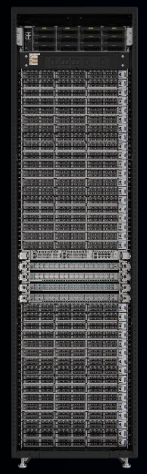
Bandwidth surges
Copper nears its limits at 224G+

Less copper, more optics
CPO

Optics + copper	Default in 3–5 years, with optics for long reach and bandwidth
CPO	From "copper traces" to "photons running in waveguides."
MicroLED CPO	Optical interconnects for scale-up networks
224G orthogonal architecture	Hybrid copper and optical interconnects within the same rack

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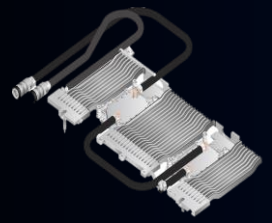
100% DLC for kW+ Chips and MW+ Racks



7

CPU & DIMM DLC

1U



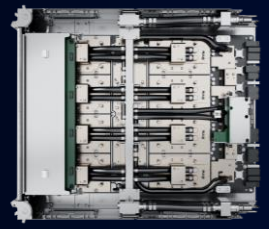
20 kW/rack
350 W/chip

2023



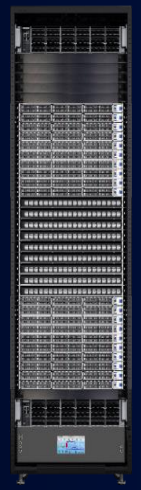
CPU & xPU DLC

4U



50 kW/rack
700 W/chip

2024



CPU & xPU DLC

1U



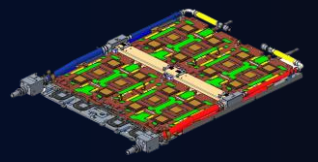
120 kW/rack
1000+ W/chip
50–100 W/cm²

2025



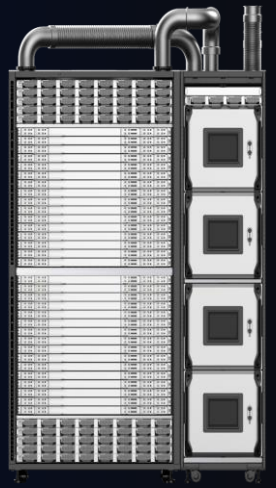
100% DLC

1U



240 kW/rack
2000+ W/chip
120 W/cm²

2026



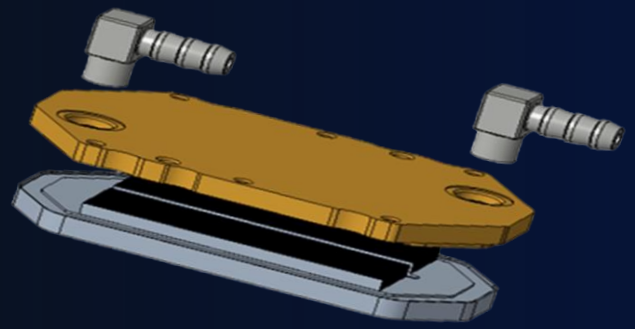
100% DLC

1U dual-width

MW/rack
3000+W/chip
200+W/cm²

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DLC Solution Meeting High Power Density AI Chips



4 kW + @200 W/cm^2 cold plate

2X

W/m*K

Diamond Cooper Application

- ≥ 650+ W/m*K thermal conductivity
- Reliability test and thermal shock

4+

kW

Higher Thermal Performance

- 4 kW+/chip thermal performance
- Low resistance micro-channel design

400+

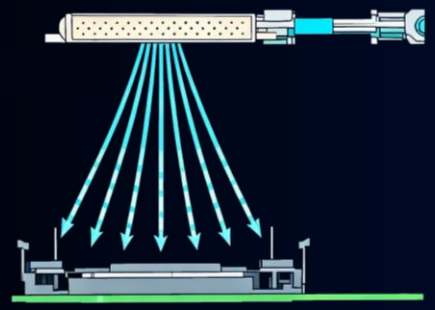
W/cm^2

Higher Heat Flux Resolved

- Non-uniform fin based on hot spot
- Average 200 W/cm^2, with 400+ W/cm^2 hot spot

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Research for the Next Generation



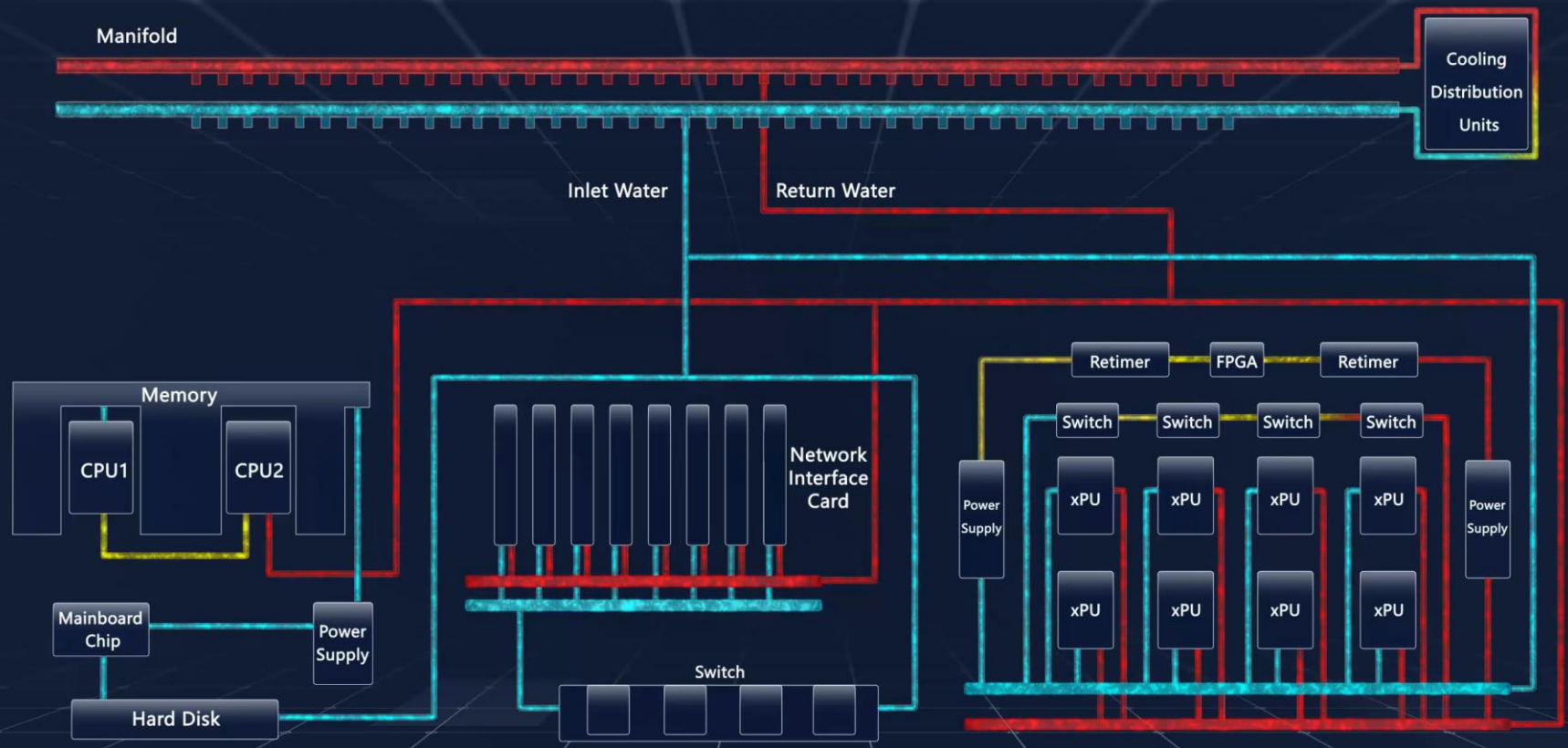
Two-phase jet cooling for chip



Two-phase cooling system

- **Enhanced two-phase jet cooling technology**
  - 130 W/m*K TIM compatible with coolant
  - Wick enhanced boiling, and jet enhanced heat transfer
- **Lower coolant cost and low pressure design**
  - Liquid level & flow control with lower cost & weight
  - zero-pressure design for stable condensing
- **Sealing solution for large dimension**
  - Flexible molecular chain, bimodal working medium sealing, ring topology, and integrated wall-through sealing
- **Rapid deployment and O&M technologies**
  - Three-bus blind mating, self-locking tool wrench, 200 kg straight-in heavy-load guide rail, and node insertion, removal, and maintenance by robots

Liquid Cooling Solution Guideline



Graphene thermal pad

The graphene thermal pad delivers thermal conductivity of 130 W/m·K with a thickness of 0.3–5 mm (0.01–0.2 in.), and completely fills irregular gaps between chips and cold plates.

Liquid metal TIM

The gallium-based liquid metal delivers thermal conductivity of 30 W/m·K, offers interfacial thermal resistance as low as 0.02°C·cm²/W, and completely fills microscopic gaps between chips and cold plates.

High-density copper shovel-tooth cold plate

The integrated copper shovel-tooth cold plate delivers thermal conductivity of 385–400 W/m·K and serves as a core heat-dissipation component in chip liquid-cooling systems.

Diamond-copper cold plate

The diamond-copper cold plate delivers thermal conductivity of 550–800 W/m·K, combining diamond's high thermal conductivity with copper's machinability for high-power chip cooling.

Diamond cold plate

The diamond cold plate delivers thermal conductivity of 800–2500 W/m·K, offering a low CTE and excellent heat-spreading performance for high power density chips.

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Power System Transformation in the Megawatt Era

From 40 kW to 1 MW, from AC to 800 V HVDC, and from silicon to wide-bandgap semiconductors, computing growth is reshaping data center power.

25x
Rack power growth over 5 years
Hopper → Feynman

800 V
HVDC: the new standard
for next-gen data centers

>97%
PSU efficiency:
beyond titanium benchmark

MW-Class
Power supply per rack
Sidecar architecture

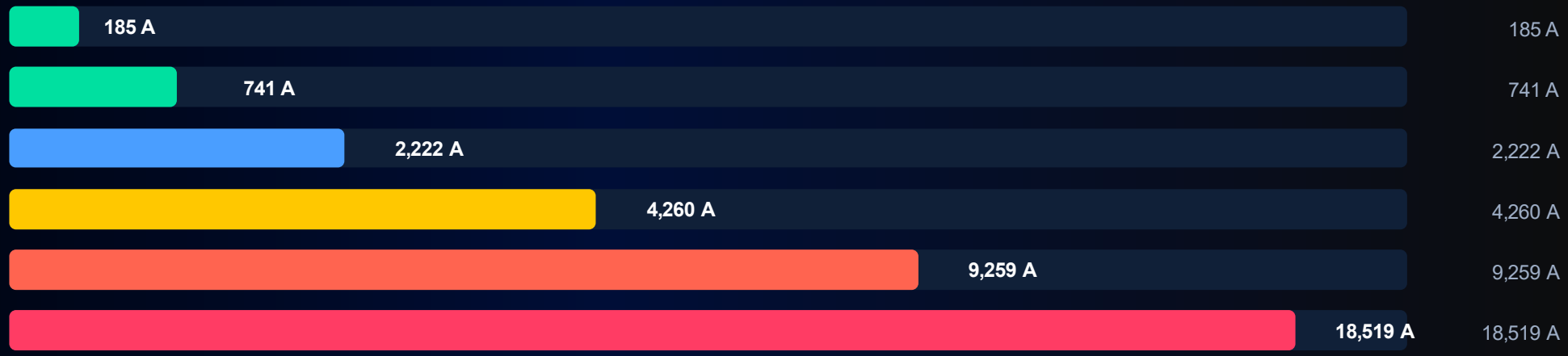
Voltage increase: from 48 V to 800 V, conversion levels: from 4 to 3, system efficiency: from 88% to 90%, and line loss: from 5%–8% to less than 1%

Voltage Level	220 V AC/ 48 V	800 V HVDC Sidecar	800 V HVDC Direct
Power	30–300 kW	100 kW–1 MW	1 MW+
Conversion levels	4 levels (AC→DC→DC→VRM)	4 levels (AC→HVDC→DC→VRM)	HVDC→DC→VRM
Conversion efficiency	97.5%/88%/92%	97.5%/88%/92%	98%/92%
Current	High current (kA-class)	Medium current (100 A-class)	Medium current (100 A-class)
Line loss ratio	5%–8%	<1%	<1%
Green power	Not supported	Limited support	Native support
Computing-power synergy	Not supported	Basic synergy	In-depth synergy

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Challenges for Power Supply in the Megawatt Era

48 V Bus Current by Rack



Limited Power Density

48 V/54 V architectures suffer quadratic copper loss at high current.

Busbar weight **200 kg+**

Space and cooling pressure

High Efficiency Loss

Six-stage DC conversion limits end-to-end efficiency to 85%–89%.

Efficiency loss **11%–15%**

Too much waste heat

Grid Access Pressure

Hyperscale AI data centers demand hundreds of megawatts from the grid.

Data center demand **100–500 MW**

Long expansion cycles and high cost

Tighter Regulation

PUE requirements are tightening to below 1.25.

Average PUE **1.46**

Challenges for efficiency compliance

- Power grows from 40 kW to 1 MW (25x higher). At 48 V → 25x more current + 625x higher busbar loss. Voltage upgrade from 48 V to 800 V → 16x higher voltage → Current drops to 1/16 → Loss drops to just 1/256.
- Every time power doubles, 48 V bus current doubles, I2R loss quadruples. At 1 MW, busbar copper loss exceeds 50 kW — one server's power wasted in every rack.

Data Source: IDC/ Gartner FCST

Power Supply Competitiveness for New Challenges

Adaptive Architecture

MW-Class Power Architecture

Power Distribution

3-phase 380 V AC PDU

800 V DC PDU

Output

PSU/BBU shelf compatibility

1.2 MW (10 x 3U powershelf);

720 kW (6 x 3U powershelf +

6 x 3U BBU shelf)

BUSBAR

Liquid/ Fan Cooling 800 V/ 1500 A

Innovative Component

Third-Gen Semiconductors

AC → HVDC

SiC HEMT →PFC+LLC →**100–300 kHz**

HVDC → DC

GaN HEMT → LLC resonant converters

> 800k-1MHz

48 V → 12 V/6 V

DrGaN HEMT→ Parallel operation →

500k–1 MHz

Vertical Power Delivery

Chip-Level VPD

Horizontal power delivery

Supplied horizontally from the cabinet side to the mainboard

Limited by PCB trace resistance and connector current capacity

Feasible up to 500 kW

↓

Dual-Side Power Delivery

Supplied from both sides of the mainboard

Lower feed resistance and higher current capacity

A solution for higher power density

↓

Vertical Power Delivery

Placed directly behind the chip (BSP/BPD)

Eliminated path resistance for ultra-low loss

A new era of MW+ single cabinet systems

Power-computing synergy

Grid interconnection & Coordination

Three-Level Energy Storage

CBU

↓

BBU

↓

Green power

Computing–Power Synergy

NVIDIA GB300 power smoothing

Infineon power pulsation buffer (PBB)

LIC+BBU

Active power buffer (APB)

Key Takeaway: Grid interconnection capacity will be the ultimate bottleneck for AI scaling. Computing–power synergy is the hidden moat of MW-class racks.

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Computing-Power Synergy Unlocks AIDC Value

68 mm (2.68 in.) High-Density PSUs

Common Redundant Power Supplies

Covering the full power range from 900 W to 3000 W



80 PLUS Titanium, CE, TÜV, CCC, NRTL, and CB certified

Power Shelf

Scalable from 2U to 8U, delivering 66 kW to 264 kW, and evolving toward MW-level systems

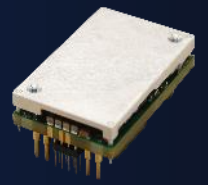


5500 W dual-input PSUs with over 97% efficiency

Board-Mount Power Supplies

Vertical Power Modules

On-board power supply with high power, over 95% efficiency, and base/cold plate cooling







UL certified, with overtemperature/overcurrent/overvoltage/short circuit protection

Broad Compatibility

Full power range from kW to MW

Full-range server power supply and distribution products and solutions, covering 900 W to MW-class applications.

10+ models, including 68 mm (2.68 in.) high-power-density PSUs, CRPSs, Power Shelf, Adapter, and Busbar

Customization support

Robust Reliability

20% lower failure rate than the industry average

1.5M+ annual shipments and 5M+ cumulative shipments, widely used in rack and rack-scale servers.

Reliability ensured by 10+ specialized tests and 1000+ test cases

200% monthly supply capacity enabled by a three-level product preparation mechanism covering materials, semi-finished products, and finished products

Optimal Energy Efficiency

Peak efficiency of up to 97%

68 mm (2.68 in.) PSUs with an industry-leading power density of 100 W/in^3

15% lower OPEX through reduced losses in core power components

Intelligent hibernation and active/standby redundancy, significantly reducing standby power consumption

Comprehensive Certification

Compliant with global standards

CE, TÜV, CCC, NRTL, and CB certifications, with PMBus 1.2 compliance

80 PLUS Platinum and Titanium certifications, meeting diverse server PSU requirements

independent and controllable solutions for various power supply and distribution scenarios

From WATTS to FLOPS: Computing-Power Synergy

End-to-End AI DC Solution Design and Delivery



xFusion

The xFusion Global Supply Center in Penang, Malaysia, is designed for an annual production capacity of 200,000 servers, encompassing xFusion's complete product portfolio.



General Computing

Up to 128 CPUs per rack
51–105 kW per rack
Hybrid air–liquid cooling

Intelligent Computing

Up to 64 CPUs per rack
105 kW per rack
Air-liquid mixing/full liquid cooling

Super Node

64/72/128 CPUs per rack
60–240 kW power supply per rack
Air-liquid mixing/full liquid cooling

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Thanks!

